Multi-evidence re-ranking for RETECO Track 1a

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Abstract

RETECO Track 1a asks for documents that answer a *complete* question about a historical period. Queries are long, their temporal and exclusion conditions sit in the body rather than the title, and the judgements mark **several** documents per question (3.44 on average) over **2.34 annotated steps**. We describe a two-stage ranker — a zero-shot dense retriever followed by a windowed cross-encoder fine-tuned on the 1,071 training queries that carry at least one positive among their dense top-100 candidates — that scores 0.3581 domain-macro nDCG@10 on the 13-domain dev set, against 0.2943 for the dense first stage and 0.1147 for the organisers' BM25 baseline. The supervised reranker contributes +0.0332 (domain-clustered 95% CI [+0.0192, +0.0472]) at a fusion form fixed before fine-tuning, and reproduces at +0.0326 / +0.0279 under two further seeds. An analysis of the frozen system shows the remaining headroom is only partly recall: at the 100 candidates the reranker sees, 80.7% of annotated steps have at least one relevant document (96.3% within the dense top-1000), yet the fused top ten covers only 59.5% of the annotated steps and the cross-encoder's own top ten 51.4%, and only 38.3% of questions place two or more relevant documents there. Explicit temporal modelling did not close this gap in ten probes, including one strictly pre-registered test (+0.0039 against a +0.02 bar) and a fixed year-consistency reordering that *lowers* nDCG@10 in 7 of 8 domains. Supervising the cross-encoder with the organisers' step annotations – so that every step must have a positive ranked high – costs 0.014 domain-macro on dev and *lowers* step coverage, 59.5% to 55.6%/56.0% in the fused list and 51.4% to 48.1%/45.8% for the cross-encoder alone. Two further corrections, a constraint-faithful 64-word full-question query and the restoration of the 140 training queries whose dense top-100 contains no positive, each cost 0.0138 on the same split - but only 0.0026 against a same-code retrain of the unchanged objective, which itself costs 0.0111, so on dev they are indistinguishable from it. Every intervention in the six-step ladder is neutral or negative on dev, and the frozen system is kept.

1 Introduction

Track 1a gives 13 independent corpora, long queries, binary qrels and no timestamps. Our entry has two parts: a strong, reproducible zero-shot-to-supervised pipeline, and a diagnosis of why the obvious temporal fixes do not work.

Contributions: (i) a baseline ladder over all 13 domains under the official domain-macro statistic, with paired bootstraps over queries and over domains; (ii) a supervised cross-encoder that adds +0.0332 without tuning the fusion weight on dev; (iii) a multi-evidence analysis showing that the system's failure mode is "one easy document" inside a candidate pool that is itself incomplete; (iv) negative evidence on explicit temporal modelling: one pre-registered test and a fixed year-consistency rule that fails; (v) a decomposition of why the cross-encoder with the best standalone score is not the one that fuses best - anchoring, attenuation and score calibration - and what that implies for using a single-stage metric to select checkpoints.

Relation to prior work. The reranker is the standard retrieve-then-rerank cross-encoder [4], initialised from M3-Embedding [12]. Our multi-evidence objective is a sibling of listwise and LambdaRank training [6, 7], with one difference: instead of ordering one list, it asks that every official step of a question own a document in the top ten. Multi-stage fusion has its own line - list-aware reranking of the first stage's list [13] and joint selection of documents that cover the answer set [14] - but both train the fusion stage end to end, whereas we freeze the first stage and ask what the second stage can add; the retrieval-side probes and the fusion-weight family below are that question. Temporal retrieval

is the benchmark's own subject [1]; our contribution there is negative evidence about what does not transfer from the two-domain pilot to the full suite.

2 Task, metric, data

Metric. Official aggregation is domain-macro nDCG@10: per-query nDCG@10, averaged within domain, then over the 13 domains, so a 3-query domain weighs as much as a 240-query one, and queries missing from a run score zero. dev = 519 queries. 26.6% of dev questions receive exactly zero from the frozen system, and 37.7% of those have no relevant document in the top 100 at all. Those zeros dominate the statistic: had the 138 zero questions merely reached half their domain's average, domain-macro would be 0.4283 instead of 0.3581 (full average: 0.4985, `results/ZERO_HEADROOM.json`).

Data. 1,167,159 documents, 1,211 train queries, binary qrels (so nDCG@10 is discounted precision at 10). Per-domain counts in `results/corpus_manifest.json`. Queries carry a title and a body separated by <p>; the body holds the temporal range, exclusions and comparisons (Section 5.3).

3 System

Dense first stage. `mxbai-embed-large-v1`, full query with the model's retrieval instruction, mean pooling, cosine similarity, fp16, `top-100` candidates per query. Dev: 0.2943 (0.2956 with CLS pooling: a wash).

Windowed cross-encoder. `bge-reranker-v2-m3` (568M). Documents are split into 250-token windows with stride 192, each window is scored against the query, and the document score is the maximum over its windows, i.e. a document is as relevant as its best passage; query capped at 256 tokens. Pooling was fixed a priori and is not ablated (Section 7). Zero-shot at fusion 1:1: 0.3249.

Fusion. Per-query z-normalisation of the dense and cross-encoder lists, combined at 1:1. The weight was fixed on the training split before any dev inspection; Section 5.1 shows the choice is not what produces the gain.

Supervised fine-tuning. Listwise softmax over a 20-document group sampled from each training query's 100 candidates - 21,420 query-document pairs in 1,071 groups, the log-sum-exp running over the group's positives - 8 of 24 layers frozen, lr 1.5e-5, seed 17, 2 epochs; epoch 1 was kept, by a 0.3029-against- 0.3012 internal-validation margin. Checkpoint selection used the cross-encoder's own ordering on 45 held-out training queries, not the fused system; that run predates domain-macro logging and selected on the plain mean, and on every later arm the two criteria pick the same epoch. Frozen dev: **0.3581**.

system	dev domain-macro nDCG@1
official BM25	0.114
dense (mxbai, top-100)	0.294
dense + zero-shot CE, z 1:1	0.324
+ SFT CE, 1 epoch	0.333
+ SFT CE, 2 epochs (frozen)	**0.358**
same recipe, seed 23	0.357
same recipe, seed 29 (two independent reruns)	0.3528 / 0.352
fresh checkpoint, RRF 1:4 (OOF-selected form)	0.318
fresh checkpoint, RRF 1:1 (V1 form)	0.343

Multi-evidence (step) supervision. Track 1a gives one relevance judgement per query, but the organisers' qrels also carry *step* annotations: a question that needs several pieces of evidence is decomposed into steps, and each step lists the documents that satisfy it. We use them only in training; at test time the ranker still sees the full question and one document. With `P_j` the step-`j` positives inside a query's candidate list, the ordinary listwise term `-log sum_{d in P} p(d)` is dominated by the single easiest positive, so we add

` L_step = -(1/m) * sum_j log sum_{d in P_j} p(d), L = L_list + lambda * L_step `

which forces *every* step to have a positive ranked high instead of letting one easy document carry the query. Steps whose positives are absent from the candidate list are skipped (a step is covered for 91.2% of step instances under our candidate construction); `m` counts the covered steps. `lambda` is not tuned on dev: two values (0.3, 1.0) are trained on the same groups, and both raise the train-internal split above the baseline (0.2882 and 0.2913 against 0.2865), yet both are rejected on dev (S5.6).

Mechanistically it targets the failure mode of S5.2: steps are retrievable (80.7% at the 100 candidates the reranker sees, 96.3% at depth 1000) far more often than they reach the top ten: 59.5% of annotated steps reach it in the fused list and 51.4% for the cross-encoder alone, with every annotated step that has a positive in the denominator.

4 Results

* the two RRF rows are compared against zero-shot CE at the *same* form (0.2981 and 0.3166), because an RRF list is not on the z-fusion scale;

system	nDCG@10	relevant docs in top-10	questions with ≥1 step	step coverage of Top-10	zero-step	nDCG@10	delta vs SFT	95% CI (domain-clu
dense	0.2943	1.03	1 step 31.8%	75	0.3185	51.3%	0.3896	+0.0711 96.3%	[+0.0244, +0
+ zero-shot CE	0.3249	1.11	2 steps 35.6%	151	0.3855	74.9%	0.4058	+0.0200 96.3%	[-0.0033, +0
+ SFT CE (frozen)	0.3581	1.21	3+ steps 28.3%	258	0.3195	59.5%	0.3511	+0.0315 96.3%	[+0.0084, +0
			none	35	0.1579		0.1705	+0.0126	[+0.0030, +0

the 0.3249 reference does not apply to them. See Section 5.1.

Robustness: the gain holds when the three smallest domains are dropped (0.3267 vs 0.2985 on n≥19 domains) and in 11 of 13 domains (the two exceptions, bitcoin and quant, are -0.0023 and -0.0009, i.e. flat). Seed spread over three seeds is 0.0029, an order of magnitude below the effect; a fourth run of the same recipe through the current code path scores 0.3470, so the recipe's retraining band is 0.347–0.358 and the frozen artifact sits at its top (S5.5, S7).

5 Analysis

5.1 Is the gain an artefact of the fusion weight?

No. A sweep over nine weights (0.5…2, CE weight) is positive everywhere (+0.0089…+0.0346; the w=1 row of that sweep reproduces the headline within bootstrap noise), and refitting the weight out-of-fold on 45 held-out training queries selects a dense-heavy RRF 1:4 that is *worse* than z 1:1, so the gain is not bought by tuning on dev.

5.2 The bottleneck is multi-evidence coverage over an incomplete pool

We annotate-free measured, per system, how much of the official step structure reaches the top ten (`scripts/dev_multievidence.py`, `results/DEV_MULTIEVIDENCE.md`):

Supervision improves every column but leaves most of the structure unused. At depth 1000, seven of ten steps are retrievable yet four of ten never reach the top ten; at the depth the reranker actually sees - the rerank files keep 100 candidates per query - only 80.7% of annotated steps are reachable and 88.3% of questions keep at least one positive, so a fifth of the step structure is lost before the cross-encoder runs. Truncating the fused list from 1000 to 100 rows leaves its top ten unchanged (59.45%, `results/MULTIEVIDENCE_sft_e2bc_sub100.json`), so depth moves reachability only; the 51.4% above is the cross-encoder's own ordering over those 100 candidates, a different list from the fused one. That limit is visible in the zeros: 52 of the 138 questions the frozen system scores at zero (38%) have no relevant document in the top 100 at all (`results/DEV_ERROR_ANALYSIS.md`), and no reranker over those candidates can lift them. The retrieval floor proper is small: in 10 of the 519 dev questions (1.9%) no relevant document appears anywhere in the dense top-1000. Coverage is not a cosmetic statistic: on the frozen dev run, per-question coverage and nDCG@10 are strongly associated (Spearman 0.69), and questions whose steps are all covered average 0.595 nDCG@10 against 0.187 otherwise (228 vs 256 questions, `results/STEP_COVERAGE_CORR.json`). The association is partly mechanical – finding a step's documents is how a question earns credit – so we use it only as a mechanism check, not as evidence of gain; arms are still judged on dev nDCG. This is the motivation for multi-evidence (step-aware) ranking supervision; Section 5.2b shows the present objective does not close it.

5.2b Where the gain sits: not on the multi-step questions

Partitioning dev by the number of official evidence steps a question needs (`results/STEP_SPLIT.md`), the frozen system's gain over zero-shot CE is largest on single-step questions and weakest on two-step ones:

The present objective therefore buys a better *single-document* decision almost everywhere, but it does not specifically teach evidence assembly: on the 151 two-step questions the gain is within noise. That is exactly the gap `L_step` targets, and it fixes the bar the step arms have to clear – a gain that moves the two-step partition, not only the overall mean.

5.3 Query truncation drops the conditions

In 1,730 train+dev queries, 44.8% (46.4% of dev: 241 of 519) have at least one condition (year 14.3%, temporal relation 19.0%, exclusion 16.3%, comparison 14.0%) that appears **only in the body**, i.e. it never reaches the cross-encoder, which is trained on the title alone; the dense stage reads the full question.

That truncation is not, by itself, what the

ranker is missing. Splitting dev by this flag (`results/CONSTRAINT_SUBSET.md`), the frozen system scores **0.3656 on the 241 condition-lost queries vs 0.3518 on the rest**, and its gain over zero-shot CE is about twice as large there (+0.0481 vs +0.0246) although the dense stage reads the full question and only the reranker does not. A same-budget body-visible composer therefore has to *earn* the claim: we pre-register that its gain must concentrate on the lost subset and not come from the kept one. Two 64-word composers were built - a cue-sentence selector that keeps the title plus only those body sentences carrying a condition, and the plain full question (title plus the first 64 body words). On the 45-query internal split no representation wins outside noise: title 0.2827, full 0.2869, cue-sentence selector 0.2744, fixed body 0.2703, frequency summary 0.2614 (`results/QVARIANT_WINNER.json`); the ~+-0.03 resolution of that split (S7) makes dev the decisive test. The selector is the more faithful of the two and it was the worst of the five, so dev carries the *generous* arm - the whole body is there and no selection step can hide a mistake. Trained end to end on that 64-word full question, it scores **0.3443** on dev against 0.3581 for the frozen system, whose reranker reads titles only (Delta -0.0138, query-clustered 95% CI [-0.0272, -0.0005], domain-clustered [-0.0338, +0.0052]), and it loses on both subsets of the pre-registered split: 0.3566 vs 0.3656 on the 241 condition-lost queries and 0.3296 vs 0.3518 on the remaining 278. The loss is thus *smaller* where the arm was designed to help (-0.0090 vs -0.0222), which is consistent with the reading that body conditions are not what the ranker is missing, but no subset gains, so the pre-registered bar (gain concentrated on the lost subset) is not met (`results/CONSTRAINT_SUBSET_cx_qv.json`).

5.4 Temporal modelling did not help

Ten probes: year/anchor features, a semantic time-consistency score, oracle and LLM step decomposition, HyDE, temporal hard negatives, a time-aware constraint rerank, a train-fitted gate, a fixed year-consistency reordering, and a coverage diagnostic. Exactly one was strictly pre-registered: it gained +0.0039 ([-0.0023, +0.0103], domain [+0.0006, +0.0142], P=0.893) against a +0.02 bar. The year rule fails cleanly: given the query's interval, ordering the frozen candidates by year consistency lowers nDCG@10 in 7 of 8 anchored domains (macro 0.2980 → 0.2498). This is a fixed rule losing, not an upper bound: the reordered list is not the best possible one, and the rule itself is what a temporal ranker would have to beat. In this corpus a document that mentions another period is often still the right answer.

objective / sampler	best epoch	val domain-macro nDCG@
current (log-sum-exp over positives)	2	0.28
mean over positives	2	0.28
nDCG-weighted permutations	2	0.27
current + sqrt(domain) sampling	2	0.29
zero-shot reference	-	0.27

5.5 The loss form is not the bottleneck

Three alternatives suggested by the coverage gap - averaging the log-loss over all positives, an nDCG-weighted permutation loss (LambdaLoss-style, cut-off 10), and domain-balanced (sqrt) sampling - were trained on the same 942 groups with the same hyper-parameters and scored on the same 45-query split:

Selecting on domain-macro rather than the plain mean picks the same epoch in all four arms, so the official aggregation does not move the checkpoint; the internal split cannot resolve the gaps themselves (+-0.03, S7), so only sqrt sampling's +0.0047 ([-0.0164, +0.0285]) is even measurable, and the 0.010 tie band fixed before seeing any number keeps the current objective. Sqrt sampling's one visible effect is raising the share of queries with a relevant document in the top ten from 93.3% to 95.6% without moving nDCG: it makes the easy decision easier, not the hard one right.

Dev separates two effects this screen conflates. The incumbent objective, retrained through the current code path (same 942 groups, two epochs, each run taking its own val-preferred epoch - the frozen run kept epoch 1, this one epoch 2), scores **0.3470**, 0.0111 below the frozen artifact (domain-clustered CI [-0.0265, +0.0057], 132 better / 167 worse). That gap is the price of re-running the recipe, not of a new objective, and it is the baseline against which the four arms above should be read. They stay unmeasured on dev: the screen separates them by at most 0.015, below the dev half-width of 0.007-0.014, and we do not carry a sign across splits. What dev supports is narrower than "the replacements lose": the screen ranks arms at 0.2764-0.2912, it cannot resolve a 0.003 difference, and the one arm it put on dev - the incumbent itself, mid-screen at 0.2865 - reproduces 0.0111

system	dev domain-macro	Delta vs frozen	domain-clustered 95% CI	steps in top 10	dev domain-micro	Delta vs frozen
frozen (no step term)	0.3581	-	frozen	59.45%	0.3581	38.31%
L_step, lambda = 0.3	0.3442	-0.0139	full-question + query-only	55.58%	0.3443	37.05%
L_step, lambda = 1.0	0.3438	-0.0143	+ 1[40 backfilled queries, 294 positives	56.00%	0.3443	37.83%

below the frozen number.

5.6 Forcing every evidence step into the top ten does not pay

The remedy the coverage gap points to is mechanical: if the fused top ten covers only 59.5% of annotated steps (51.4% for the cross-encoder alone), require every step to have a positive ranked high. We trained the step-augmented listwise objective of S3 at two weights and scored the resulting systems on dev with the frozen fusion:

The coverage column counts the fused list (`runs/sub_*`); its top ten does not change when the list is truncated from 1000 to 100 rows (59.45%, `results/MULTIEVIDENCE_sft_e2bc_sub100.json`). The cross-encoder's own ordering over the same 100 candidates covers 51.39% before the step term and 48.14% / 45.78% with it (`results/MULTIEVIDENCE_*_depth100.json`), so the degradation is a property of the ordering, not of the candidate depth.

The term does not merely fail to pay for itself: it *reduces* the quantity it optimises. Both arms also lose on the population they target, the two-step questions (0.4058 to 0.3910 and 0.3889, CI including zero), and the loss is spread over all partitions. Two readings fit: a step's positives are mostly documents the ordinary listwise term already covers (3.44 relevant documents over 2.34 steps, so a step rarely adds a *different* document), and the log-sum-exp inside a step is dominated by the step whose positive is hardest to retrieve, so the gradient chases documents that are absent or buried. The annotations are not at fault: every one of the 2,873 step-level positives is also a query-level positive.

The instructive part is the selection behaviour. On the 45-query internal split lambda = 1.0 was the *best* of all six arms trained that day (0.2913, above the current objective's 0.2865), and dev puts it last. At this size the internal ranking is not a weak signal but the wrong one, which is why S3 fixes the fusion form in advance and S7 refuses to select anything on 45 questions.

5.7 Two further corrections that also lose

The frozen system feeds the reranker a title-only query and excludes the 140 training queries whose dense top-100 contains no positive. Both were corrected with the frozen recipe (two epochs, same schedule) and scored on dev with the frozen fusion:

The two corrections cost exactly the same 0.0138, which suggests a flat basin in both directions rather than two intrinsically bad ideas. The mechanisms differ. For the query arm the loss is smaller where the body conditions live (S5.3), so the extra text carries signal but not enough to pay for the harm on the other 278 queries; the reranker is better at its job with a short, clean query. For the backfill arm the 294 restored positives are, by construction, documents the first stage never retrieves, so training on them spends capacity on a decision never made at inference, where the candidate list is the dense top-100.

The selection behaviour repeated itself: the full-question arm's 0.2920 was the highest of the seven arms trained on 11-10 on the internal split, and it lands 0.0138 below the system it was meant to improve. Most of that is not the correction. The unchanged objective re-run through the current code path costs 0.0111 (S5.5), and against that same-code baseline the two corrections cost 0.0026 each, with domain-clustered intervals [-0.0187, +0.0100] and [-0.0162, +0.0101] that include zero; the two step arms behave the same way (-0.0027 [-0.0154, +0.0082] and -0.0031 [-0.0153, +0.0072], `results/ATTRIBUTION_LXPOSSUM.json`). Dev therefore supports a deployment decision - five of the ladder's six steps were retrained and none beats the frozen 0.3581, and the sixth (dense pooling) is a wash at +0.0014 - plus a negative result against a same-code baseline that the 45-query screen cannot rank at all.

5.8 A better cross-encoder is not a better system

The checkpoint screen scores the cross-encoder on its own validation nDCG; the submitted system is the dense+CE fusion, and the two do not rank the arms the same way. On dev `cx_qv` is the stronger cross-encoder by 0.0397 (0.3153 against 0.2756) and 0.0027 *worse* after fusion at the frozen weight;

arm	cross-encoder alone	fused, w = 1	fused, best w
frozen recipe (`lx_possum`)	0.2756	0.3470	0.3470 (w = 1)
full-question (`cx_qv`)	0.3153	0.3443	0.3546 (w = 2)

across the six arms scored, the best CE-only arm is third fused and the second-best is first (Spearman 0.66).

Three mechanisms explain the loss (519 dev queries, frozen dense top-1000, per-query z-normalised 1:1 fusion). *(i) Anchoring.* The fused top ten is 74% dense documents and only 55-60% of it is also in the CE top ten, so a changed ordering must displace dense documents to matter. *(ii) Attenuation.* A CE gain transfers at a slope of 0.30: +0.1985 on the 185 queries the CE improves becomes +0.0458 fused, and -0.1398 on the 113 it hurts becomes -0.0482. *(iii) Calibration.* The score profile matters more than the ordering: on the 221 queries where the CE's own nDCG@10 does not move (|deltal <= 0.01), the fused score still falls 0.0116. A transplant isolates the two: `cx_qv`'s ordering with `lx_possum`'s profile scores 0.3482 (+0.0012 over `lx_possum`), while `lx_possum`'s ordering with `cx_qv`'s profile falls to 0.3403 (-0.0067). At w = 1 the fused list follows the score profile more than the better ordering.

Re-weighting is not a free fix. The optimum is model-specific (`cx_qv` wants w = 2, worth +0.0103 [+0.0012, +0.0201] over w = 1, recovering about a quarter of the CE gain), but the weight must be chosen somewhere: chosen on the 45 held-out queries (argmax 1.25) `cx_qv` reaches 0.3509, still -0.0072 against the frozen system, and the same procedure costs `lx_possum` 0.0058 ([-0.0117, -0.0003]). A calibration-free RRF (k = 60) recovers +0.0069 over the same weight and also trails the frozen system, by 0.0119.

On the same 45 queries the inversion reproduces with the candidate list fixed: `cx_qv` is 0.043 better alone (0.2893 against 0.2462) and 0.013 worse fused (0.3197 against 0.3325). Within either model both criteria prefer epoch 1 - and that preference is itself noise: re-scoring `lx_possum`'s epoch 1 on dev gives 0.3428 against 0.3470 for the epoch the original screen kept (domain CI [-0.0114, +0.0022], 397 of 519 topics tied). On 45 queries neither criterion resolves a 0.004 difference. What we can attribute is the *model*-level inversion, not an epoch-level mistake: a single-stage metric is a good ranker of cross-encoders and a poor ranker of the system they feed, because the fusion re-normalises both scores per query and the first stage dominates the list.

6 Reproducibility and efficiency

Every artefact is content-addressed: document embeddings, dense runs, CE runs and submissions carry a fingerprint sidecar (`.fp.json`) that binds model, pooling, max length, query file and fusion parameters, and the submission writer refuses to reuse a stale cache. The submission path was replayed end to end on synthetic test splits and passes the organisers' `format_checker.py` with zero errors on three domains. Seed ensembling of the cross-encoder was tried and rejected: z-averaging the three seeds before fusion gives 0.3567, i.e. -0.0014 against the single checkpoint (domain-clustered CI [-0.0032, +0.0001], 481 of 519 topics tied), so the submission ships one model (`results/ENSEMBLE.md`). Two CE epochs take 55 minutes and dev re-ranking costs ≈34 GPU-minutes for 519 queries, both on one 8 GB laptop GPU; the dense document embeddings are the only large artefact (several GB on disk).

7 Limitations

Domain-macro lets 3-query domains weigh as much as 240-query ones; the temporal findings are about *textual* time only, not about systems with timestamps; supervised results are single-corpus and single-family (one dense model, one cross-encoder). The window/pooling schedule (250-token windows, stride 192, max over windows, 256-token query cap) was fixed a priori and is not ablated here; the dense encoder's pooling was ablated late and is a wash (CLS 0.2956 against 0.2943 for mean); the 1:1 fusion weight and the two-epoch schedule were picked on the training-internal split, so the frozen number carries that selection. The step-supervision results are reported at one seed; a second seed, and distilling a larger re-ranker into the 568M encoder (Hinton et al., 2015), were not affordable within the GPU budget of this submission.

Selection is noisy by construction: the 45-query internal split improves the sum-positive arm by +0.0135 with a domain-clustered interval of [-0.0317, +0.0571], i.e. it cannot resolve differences below about 0.04. Every arm here is therefore screened internally and judged on dev with a paired domain bootstrap (half-width 0.007-0.014); dev gains below +0.010 are treated as unresolved. The

failure	query	gold	dense rank	CE rank	fused rank	nDCG@10
pool miss	travel/201562_1094	2	656 / not retrieved	1	1	0.1000
ordering	history/70583_271	1	24	14	17	0.0001
step coverage	history/59309_621	2	1, 10	1, 72	1, 20	0.6131

screen's record is not that it promoted every loser: one arm was its top score (the full-question arm, 0.2920), the loss sweep's winner fell inside the pre-registered 0.010 tie band and the base was kept (`LOSS_WINNER.json`), stage 3 kept the same base (`STAGE3_WINNER.json`), one arm was trained to fix a documented defect rather than because it won, and the step arms were run as a pair. It is rather that the screen cannot rank them: the four arms it scored on the shared split lie within 0.006 of each other and dev puts them 0.0026-0.0031 apart from the same-code baseline, all with intervals that include zero.

Neither is retraining free: the frozen recipe re-run under three seeds gives 0.3575 / 0.3528 / 0.3527 on dev with freshly sampled pools (14.6-14.8k history documents against 12.5k in the frozen run), and a fourth run through the current code path 0.3470. The retraining band is therefore 0.347-0.358, and 0.3581 is the property of the submitted artifact rather than a number the recipe reproduces on demand; we did not hold negative-pool sampling fixed.

8 Error analysis

A quarter of dev questions (138/519, 26.6%) score exactly zero. The split separates cleanly by candidate-pool recall: **52 (37.7%) have no relevant document in the dense top-100** - no top-100 reranker can score them - and 86 (62.3%) have the gold in the pool and lose it to ordering. Per domain the zeros cluster where the corpus is thin (history 46, politics 14, bitcoin 13) and are absent where it is dense (iota 0). Three worked examples, one per mechanism (`results/DEV_ERROR_ANALYSIS.md`):

The pool-miss case is a PNR booking-limit question whose evidence sits outside the 1,000 documents the dense stage returns - the answer is not near the lexical neighbourhood of the question, and no reranking depth recovers it. The ordering case ("How do we know when Nero was born?") has its single gold document at dense rank 24; the cross-encoder moves it to 14, fusion to 17, and the question scores zero. The step case ("How many people watched the moon landings after Apollo 11?") is that the multi-evidence metric is about: the Apollo 11 figures are rank 1 everywhere, the comparison document sits at dense rank 10 and falls to 72 for the cross-encoder - a short title-only query retains the entity but loses the comparison, so the fused list keeps one evidence document out of two and nDCG@10 stays high (0.61) while step coverage fails. This is the common shape, not an outlier: 38.3% of questions place two or more relevant documents in the top ten (S5.2).

9 Conclusion

A zero-shot dense first stage plus a cross-encoder fine-tuned on the 1,071 questions with an in-pool positive gives +0.0332 domain-macro nDCG@10 over the dense baseline on all 13 dev domains, and the gain is stable across seeds (0.347-0.358) and fusion forms. The remaining headroom is not one failure but two: an incomplete candidate pool (1.9% of questions have no gold in the dense top-1,000, and 38% of zero-scoring questions have none in the top 100) and multi-evidence coverage (59.5% of annotated steps in the fused top ten). Every explicit temporal intervention tested - ten retrieval-side probes, one pre-registered test, step supervision, query reformulation, loss variants, backfilling and rank fusion - is neutral or negative on dev, and the frozen system is kept. The most transferable finding is the last one: a cross-encoder that is 0.04 better on its own metric can be worse in the fused system, because fusion anchors on the first stage and re-normalises the second stage's scores per query, so single-stage validation is a weak checkpoint criterion.

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